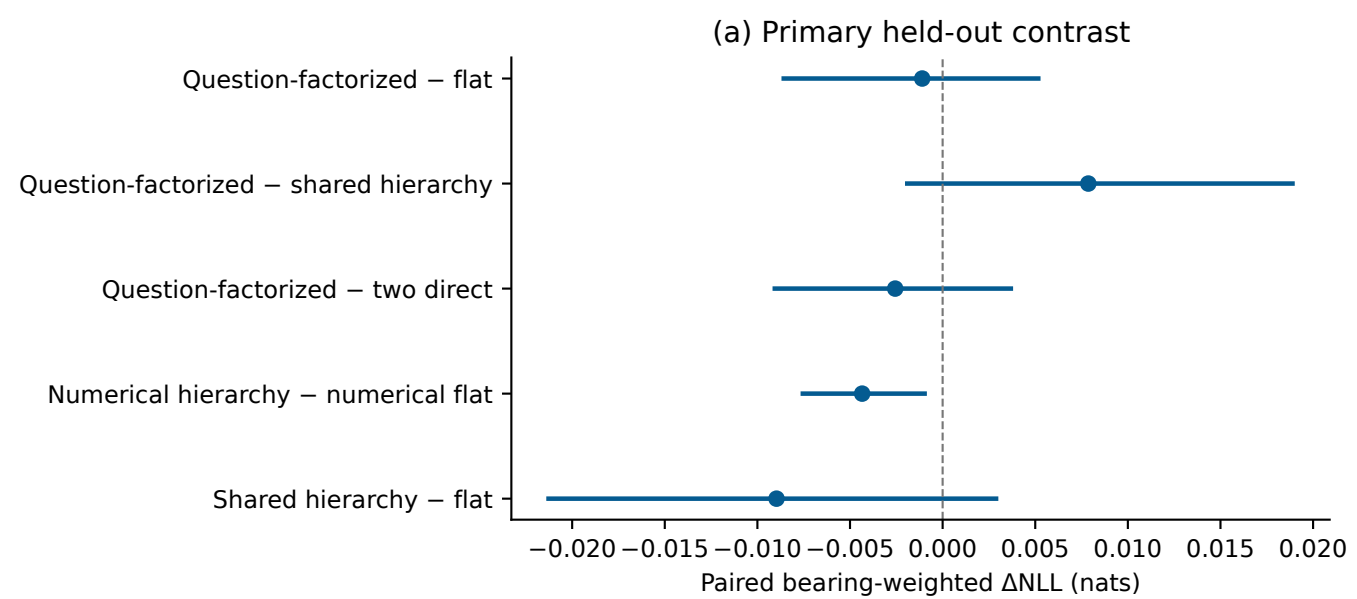
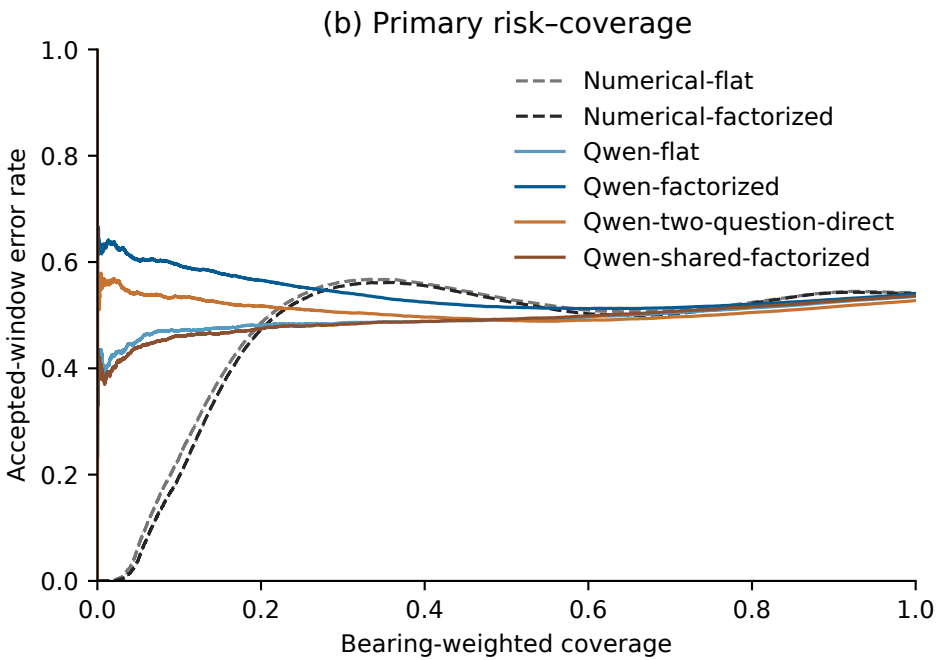

(a) Primary held-out contrast
Question-factorized − flat
Question-factorized − shared hierarchy
Question-factorized − two direct
Numerical hierarchy − numerical flat
Shared hierarchy − flat
−0.020
−0.015
−0.010
−0.005
0.000
0.005
0.010
0.015
0.020
Paired bearing-weighted ΔNLL (nats)


(b) Primary risk–coverage
Numerical-flat
Numerical-factorized
Qwen-flat
Qwen-factorized
Qwen-two-question-direct
Qwen-shared-factorized
Accepted-window error rate
0.0
0.2
0.4
0.6
0.8
1.0
Bearing-weighted coverage